SAI VISHNU VAMSI SENAGASETTY

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SUMMARY

AI Developer who ships production RAG and LLM systems for client-facing platforms end to end: gathering requirements directly from clients, building the pipeline, and owning it after deployment. Delivered measurable results across chatbot, voice, and backend-integration work, including double-digit gains in response quality and major cuts to operating cost and manual turnaround time.

WORK EXPERIENCE

AI Developer — Zibtek (Full-Time Intern) **Apr 2026 – Present**

- Lead AI development for a client production chatbot, cutting ungrounded responses by ~45% and lifting answer relevance by ~35%.
- Built an end-to-end TTS narration pipeline (OpenAI gpt-4o-mini-tts) that batch-generates narration audio across 27+ pages, cutting content turnaround time by ~60%.
- Build core backend integrations for enterprise client platforms using Node.js and Express; use Cursor to accelerate CRUD/API development, automated test-suite generation, PDF/report-generation logic, database migrations, and debugging.
- Gather requirements directly from US-based clients, conduct technical interviews for engineering hires, and co-author PRDs with team leads and senior developers to scope project estimates.
- Built a Python/FastAPI AI service layer using PyTorch and Hugging Face Transformers for model inference, with spaCy preprocessing and FAISS-based vector retrieval to ground LLM responses in domain-specific content.

AI Software Engineer — boomerroomers LLC **Jan 2026 – Present**

- Designed and deployed a multi-tier AI deflection system with a 5-stage response funnel and semantic caching, automating resolution of 70% of user queries at zero marginal cost for a global housing platform.
- Engineered a scalable RAG pipeline using Llama-3.3-70B (Groq LPU) and Gemini embeddings, cutting operational costs by 98% while delivering low-latency, context-aware responses via real-time document retrieval.
- Built a dynamic “Guided Discovery” onboarding agent that improved onboarding completion by 30%.
- Developed AI governance infrastructure, including 768-dimension vector search (MongoDB Atlas), regex-based PII masking, and a hallucination-monitoring framework, improving retrieval precision and response reliability.

AI Researcher Fellow — Handshake (Remote) **Dec 2025 – Present**

- Construct red-teaming prompt structures to probe model boundaries, identifying failure modes in negative constraints and spatial reasoning that automated metrics (CLIP, FID) failed to detect.
- Evaluate LLM code-generation performance via end-to-end web-application stress tests (e.g., Kanban-style systems), producing ground-truth datasets that support iterative model training.

ADDITIONAL EXPERIENCE

Instructional Assistant – Web Development & CMS, University of Houston — Sep 2024 – Feb 2026

AI Research Intern, Grandios Mobility (Remote) — Aug 2024 – Dec 2024

NLP Research Assistant, University of Houston — Jan 2024 – Jul 2024

PROJECTS

AI Portfolio Platform — RAG Chatbot + LLM Job-Fit Matcher

- Built a Next.js/React platform with a deployed RAG chatbot proxy and an LLM-assisted job-fit matcher combining deterministic skill-taxonomy scoring with Groq-based pitch rewriting.

Real-Time Meeting Copilot (Whisper + LLM Suggestion Engine)

- Built a FastAPI service using Groq Whisper for streaming transcription (NDJSON) and schema-locked LLM suggestion generation across multiple context modes, paired with a Next.js three-pane UI.

Automated Document Intelligence Pipeline (FAISS + Groq RAG)

- Built a 3-service Dockerized pipeline (parse → FAISS retrieval + Groq analysis → report generation) orchestrated by Apache Airflow, with CI/CD publishing images via GitHub Actions.

SKILLS

AI/LLM: RAG, Agentic AI, LangChain, FAISS, Prompt Engineering, LLM Fine-Tuning, Model Training, NLP, Computer Vision, Object Detection, Vector Databases, Hallucination Mitigation, RLHF/DPO

Languages/Frameworks: Python, JavaScript/TypeScript, Node.js, SQL, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, FastAPI, Flask, Next.js, React, Hugging Face

Infrastructure: Docker, Kubernetes, DevOps, Apache Airflow, AWS, Azure ML, MongoDB, PostgreSQL, Redis, Git, CI/CD

PUBLICATIONS

Exposing Pink Slime Journalism: Linguistic Signatures and Robust Detection Against LLM-Generated Threats — arxiv.org/abs/2512.05331

SOH of Lithium-ion Batteries by Data-Driven Technique with Optimized GPR — ieeexplore.ieee.org/document/10169188

A Night-Patrolling Mechanism Using Deep Learning — igi-global.com/gateway/chapter/318080

EDUCATION

M.S. in Computer Science, University of Houston — Aug 2023 – May 2025

B.Tech in Computer Science, SRM University, AP — Jun 2019 – May 2023

Certifications: OCI 2025 Certified Generative AI Professional • Machine Learning A-Z (Python & R) • Data Manipulation in SQL